Doohickey bluepaper

Sirusdog

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1 Basic installation

1.1 Initial setup

1. Flash an SD card with Rasbian Lite using the raspberry pi imager. Keep in mind that this code should work on other hardware/os versions, but setup will likely differ.
2. Plug in the RPI, connect it to a display via HDMI and give it power via the USB-C port.
3. I'd reccommend leaving the username as pi and setting a sudo password you'll remember.
4. Connect the RPI to the internet, either via it's ethernet port or WIFI.

1.2 Software installs and interface setup

1. Run *sudo apt install git*
2. Run *git clone https://github.com/Sirusdog/Thermal-camera-project.git*
3. Run *cd Thermal-Camera-Project*
4. Run *sudo raspi-config*
   (a) Select *3 Interface Options*
   (b) Enable *I4 SPI*
   (c) Enable *I5 I2C*
   (d) Select *I6 Serial port*
      i. Select *No* for "Would you like a login shell to be accessible over serial?"
      ii. Select *Yes* for "Would you like the serial port hardware to be enabled?"
   (e) Optionally you can disable the sudo password and enable auto-login to root. I'd also reccomend enabling SSH.
   (f) Apply the settings.
5. Run *sudo bash installer.sh*
6. When prompted to reboot press Y.

1.3 Setting up composite display

1. Run *sudo raspi-config*
2. Select *2 Display Options*
3. Select *D4 Composite*
4. Select *Yes* for ”Would you like composite video output to be enabled?”
5. Press *Ok*, *Finish*
6. Run *sudo reboot*
7. Connect the display to the composite output.
8. Verify that it’s working before moving on. At this stage if there’s no display output you’ll need to SSH into the raspi to figure out what’s happening.

2 Manual installation

2.1 Initial setup

1. Flash an SD card with Rasbian Lite using the raspberry pi imager. Keep in mind that this code should work on other hardware/os versions, but setup will likely differ.
2. Plug in the RPI, connect it to a display via HDMI and give it power via the USB-C port.
3. Connect the RPI to the internet, either via it’s ethernet port or WIFI.

2.2 Software installs and interface setup

1. Run *sudo apt update* and *sudo apt full-upgrade*
2. Run *sudo apt install git* (i forgot if git is installed by default)
3. Run *git clone https://github.com/Sirusdog/Thermal-camera-project.git*
4. Run *cd Thermal-Camera-Project*
5. Run *pip install -r requirements.txt –break-system-packages*
6. Run *sudo raspi-config*
   (a) Select *3 Interface Options*
   (b) Enable *I4 SPI*
   (c) Enable *I5 I2C*
   (d) Select *I6 Serial port*
      i. Select *No* for ”Would you like a login shell to be accessible over serial?”
      ii. Select *Yes* for ”Would you like the serial port hardware to be enabled?”
   (e) Optionally you can disable the sudo password and enable auto-login to root. I’d also reccomend enabling SSH.
   (f) Apply the settings.
7. Run *sudo reboot*

2.3 Cythonizing main and setting up auto-run

1. Run *cd Thermal-Camera-Project*
2. Run *python3 initialiser.py*
3. Run *python setup.py build_ext –inplace*
4. Run *sudo nano thermalCamBoot.service*
   (a) Change the line *ExecStart=/user.bin/python3 /home/sirus/Thermal-camera-project/run.py* so that the path matches the username of your raspi, i.e if you set the pi's username to pi it'll be *ExecStart=/user.bin/python3 /home/pi/Thermal-camera-project/run.py*
   (b) Press *Ctrl + X* and then *Y* to save the file.
5. Run *sudo cp thermalCamBoot.service /etc/systemd/system/*
6. Run *sudo systemctl daemon-reload*
7. Run *sudo systemctl enable thermalCamBoot* TODO: Check if this starts the program.

Past this point I'd reccomend assembing the (using a normal screen and psu) to test if everything software-wise is working.

2.4 Setting up composite display

1. Run *sudo raspi-config*
2. Select *2 Display Options*
3. Select *D4 Composite*
4. Select *Yes* for "Would you like composite video output to be enabled?"
5. Press *Ok*, *Finish*
6. Run *sudo reboot*
7. Connect the display to the composite output.
8. Verify that it's working before moving on. At this stage if there's no display output you'll need to SSH into the raspi to figure out what's happening.

3 Parts

Optical elements

Item ID	Description	Designations
LB1757	F=30 mm, 25.4 mm biconvex lens	f_1
ME1-G01	25.4 mm first surface mirror	m
EBS1	25.4 mm 50:50 Beamsplitter	b
PL25mm	25mm Plossl eyepiece	f_2
LB—	F=20 mm, 12.4 mm biconvex lens	f_3

Note that prices do *not* include shipping/vendor costs.

GX16 Connector Pinout

Pin #	Name	RPI	NVG
1	5v	GPIO 5V pins	Thermal camera 5V in, display 5V in
2	GND 1	GPIO GND pin	Thermal camera GND, rotary encoder GND
3	3.3v	GPIO 3.3V pins	Rotary encoder VCC
4	Encoder CLK	GPIO 23	Rotary encoder CLK pin
5	Encoder DT	GPIO 24	Rotary encoder DT pin
6	Encoder SW	GPIO 25	Rotary encoder SW pin
7	Cam CVBS	CVBS-USB board CVBS input	Camera CVBS output
8	Cam RX	UART0 TX	Cam RX pin
9	Cam TX	UART0 RX	Cam TX pin
10	Screen CVBS	RPI composite out	Screen composite in
11	GND 2	GPIO GND pin	Sceen GND, brightness controls
12	NC	NC	NC

Part List - Normal version

Name	Qty	Cost (€)	Link
MX10130 White phosphor autogating NNVT5 tube	1	1385.39	Alibaba link (likely cheaper now)
PVS-7 collimator	1	178.46	Bought off of Ebay
PVS-7 objective lens	1	176.14	Bought off of Ebay
0.8V-5V to 3V Voltage regulator	1	6.15	Aliexpress
5mW IR Laser (Line)	1	12.46	Aliexpress
LB1757 F=30mm, D=25.4mm biconvex lens	2	28.56	Thorlabs
ME-G01 Aluminum Mirror, 3.2 mm Thick	2	33.33	Thorlabs
C&K COMPONENTS A20403RNZQ 4P2S Rotary switch	1	12.1	Farnell
Momentary switch	1	?	Lost track
25mm Orbinar Plossl Eyepiece	2	26.90	Amazon
10mm M3 nuts and bolts	20	?	Hardware store

Extended parts list - Thermal version

Name	Qty	Cost (€)	Link
Mini2 Thermal Camera	1	174.20	Ebay (Can be bought from Aliexpress)
0.39in OLED Display (AV tricolour)	1	73.23	Aliexpress
KY-040 Rotary Encoders	1	7.32	Amazon
120 Dupont wires	1	12.68	Amazon
EBS1 50:50 Beamsplitter	1	37.27	Thorlabs
LB1494 F=12mm, D=9mm biconvex lens	1	26.70	Thorlabs
GX16 12 pin connectors male	2	a	a
GX16 12 pin cable female-female	1	a	a
Raspberry Pi 4B 2GB	1	61.65	The Pi Hut
CVBS-USB converter board	1	38	Ivcan
Audio jack to RCA cable	1	5.02	The Pi Hut
Heatsink	1	0.93	The Pi Hut
1.25mm Pitch 5-pin 28AWG Cable	1	1.05	The Pi Hut

Extended parts list - GPS, compass & waypoints

Name	Qty	Cost (€)	Link
USB GPS Reciever	1	14.94	The Pi Hut
Adafruit BNO085 9-DOF IMU breakout	1	28.02	The Pi Hut